A MASTER PAPER / BEGIN WHEREVER YOU FEEL COMFORTABLE

Mathematics, one idea at a time

Jeff, this is not a test. Think of it as a walk from a shelter room to a ladder, then out to the Moon and the moving planets. We will name each idea before asking it to do any work.

The four subjects are not four locked doors. Each adds a new kind of question to the same habit: **say what you know, say what you want, and connect the two.**

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The history and science behind the stories

How to read: stop whenever an idea feels new. Read the words, look at the picture, then try the numbers. Nothing here requires formal calculus notation.

The contents and colored page references are clickable. Click any page footer to return here. Examples simplify real situations; they are not operating limits or engineering advice.

1.1 / ALGEBRA STARTS WITH AN EMPTY PLACE

How many people in the shelter?

Picture a shelter with three rooms. For this arithmetic example, imagine each room holds 40 people. How many people is that altogether?

Algebra is a way of reasoning about an unknown number. An **unknown** is simply the answer we have not found yet. We can leave an empty box for it, or give it a letter as a label.



Here, let **C** stand for the total number of people. The letter is not a special object. It is just the name on the empty place we want to fill.

The sign **×** means “times,” or “multiplied by.” The sign **=** means “has the same value as.” It is not an instruction to guess.

THREE GROUPS OF FORTY PEOPLE

C = 3 × 40 = 120 people

Check the answer: 40 people in the first room, 40 in the second and 40 in the third make 120. In symbols, **+** means “add”: **40 + 40 + 40 = 120**.

We did not make the problem harder by writing C. We made the question visible: “What total belongs here?”

The room capacity is invented for teaching. Real shelter capacity depends on space, access, staffing, safety requirements and local rules.

1.2 / WORK BACKWARD, THEN CHECK

How many pallets in the truck?

Now a truck needs to carry a 3,000-pound load. Suppose each loaded pallet weighs 500 pounds. How many pallets make that load?

Let **P** be the number of pallets. Before we know P, we can still describe the relationship: the number of pallets times the weight of each pallet gives the total weight.

NAME THE UNKNOWN

P × 500 pounds = 3,000 pounds



“How many groups of 500 fit into 3,000?” is a division question. The sign ÷ means “divided by.” Division undoes multiplication.

FILL THE EMPTY PLACE

P = 3,000 ÷ 500 = 6 pallets

Check: six pallets times 500 pounds per pallet gives 3,000 pounds. Putting our answer back into the original relationship confirms that it fits.

Dividing both sides of an equality by the same number other than zero keeps them equal. Here, dividing each side by 500 leaves **P = 6**.

Name the unknown, use the relationship, then check. We will do it again with the ladder in Section 3 (page 7).

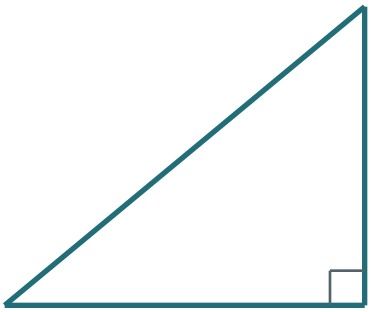
“lb” on the boxes means pounds. This is weight arithmetic only—not a loading plan. Real loads must meet vehicle, axle, space, balance and securement limits.

2.1 / GEOMETRY GIVES THE PROBLEM A SHAPE

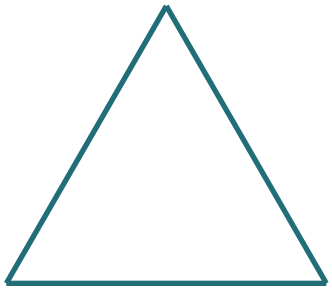
Euclid’s world of lines and corners

People measured fields and built structures long before the Greek mathematician Euclid. Around 300 BCE—about 2,300 years ago—he organized much earlier knowledge into a book called Elements.

Geometry studies shapes, sizes and positions. Euclid’s lasting example was to begin with clear assumptions and build an explanation, step by step. A **proof** is reasoning that shows why a claim must be true.



Right triangle



Equilateral triangle

An **angle** describes how widely two lines open apart. We measure angles in **degrees**, written with the little sign °. A full turn is 360 degrees; a square corner is 90 degrees.

A **right triangle** has one square corner. “Right” here means 90 degrees, not the right-hand side of the page. Its **hypotenuse** is the side across from that corner, and is always the longest side.

An **equilateral triangle** has three equal sides. On flat paper, each of its three angles is 60 degrees. It does **not** have a square corner.

These are different triangles. We can split the equal-sided one to make a right triangle; that connection will explain the ladder’s one-half relationship on the next diagram (page 6).

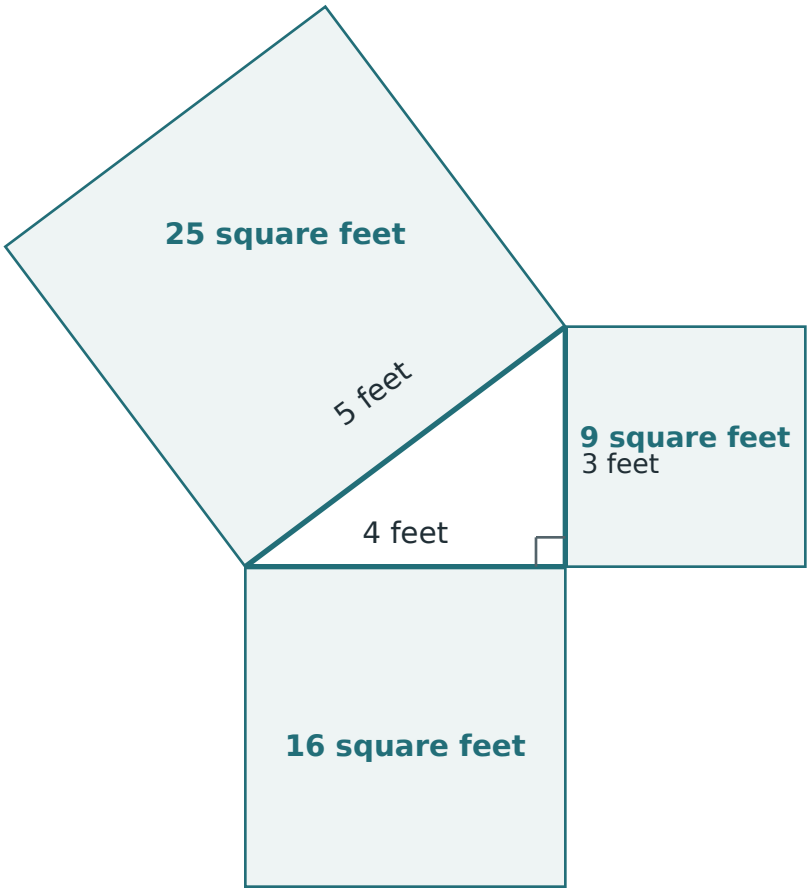
BCE means “before the common era.” Geometry is much older than any one mathematician. History sources are collected on page 22.

2.2 / PYTHAGORAS CONNECTS THREE LENGTHS

The 3–4–5 triangle

Imagine a right triangle whose two square-corner sides measure 3 feet and 4 feet. Its hypotenuse measures 5 feet. Why do these three lengths fit?

Pythagoras's theorem says that the area of the square built on the hypotenuse equals the areas of the other two squares added together. A **theorem** is a statement established by proof.



Area means how much surface is covered. A square 3 feet on each side covers 3 times 3, or 9 square feet. A **square foot** is a square one foot on each side.

So our picture says: **3 × 3 + 4 × 4 = 5 × 5**, or **9 + 16 = 25**. If the long side were unknown, we would ask: "What length times itself gives 25?" The answer is 5.

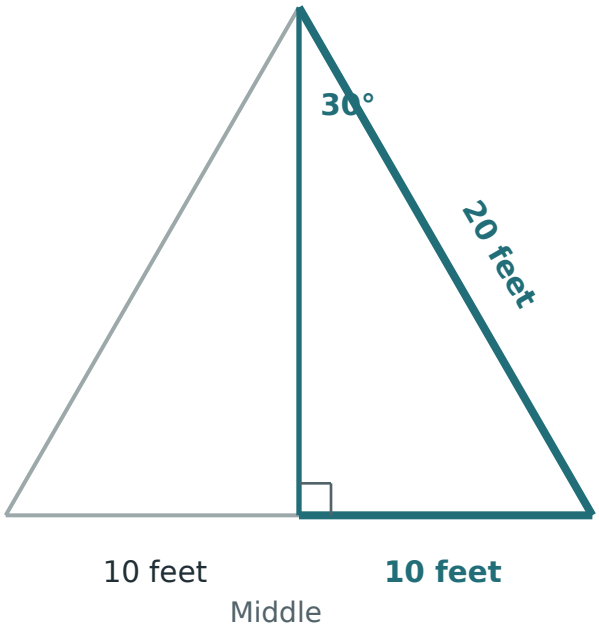
The theorem bears Pythagoras's name, but the relationship was known in Babylonian mathematics much earlier. Galileo used this kind of right-triangle reasoning (page 12).

2.3 / A BRIDGE FROM GEOMETRY TO TRIGONOMETRY

Why thirty degrees gives one half

Begin with an equilateral triangle: three equal sides, each 20 feet long. Every corner is 60 degrees. Draw a line from the top corner straight down to the middle of the bottom side.

The line splits the picture into two matching right triangles. It cuts the top angle into two 30-degree angles and the bottom side into two 10-foot pieces.



In either half, the side **opposite the 30-degree angle** is 10 feet long. The sloping side—the hypotenuse—is 20 feet long. So the opposite side is **half** the hypotenuse.

Turn that half-triangle around on the page: the lengths and angles do not change. It is the same shape as our 20-foot ladder reaching 10 feet up the wall.

A ratio compares two lengths by division. Here, 10 ÷ 20 is one half. That comparison comes from the shape, not a lucky measurement or a number invented by a calculator.

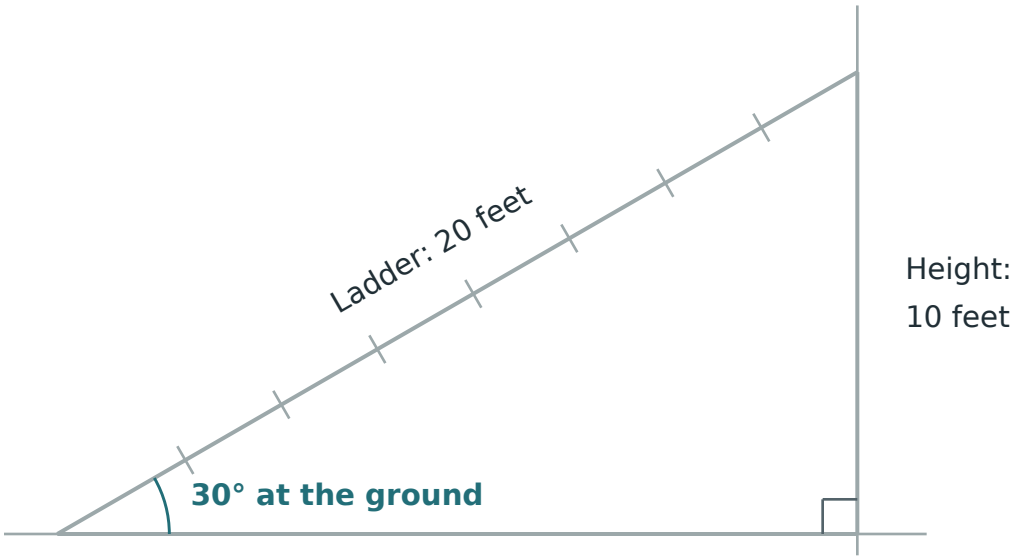
This ratio is called **sine** (Section 3, page 8). Geometry supplies the shape; trigonometry uses its proportions.

BEGIN WITH SOMETHING YOU CAN PICTURE

One ladder, three questions

Return to the right triangle from Section 2 (page 4). A ladder leaning against a wall makes that shape. We will keep the **same ladder** throughout this section.

Trigonometry connects angles with side lengths. It adds known ratios to algebra's search for an unknown (Section 1, page 2). You do not need calculus yet.



Base: about 17.32 feet

First, give the parts their names.

Right triangle: a triangle with one square corner, like a sheet of paper. Here, that corner is where the wall meets the ground.

Hypotenuse (hy-POT-uh-noos): the side across from the square corner. It is always the longest side. Here, it is the ladder.

Angle: how wide two lines open apart. We measure from the ground up to the ladder: **30 degrees**. The little ° sign means "degrees." A square corner is 90 degrees.

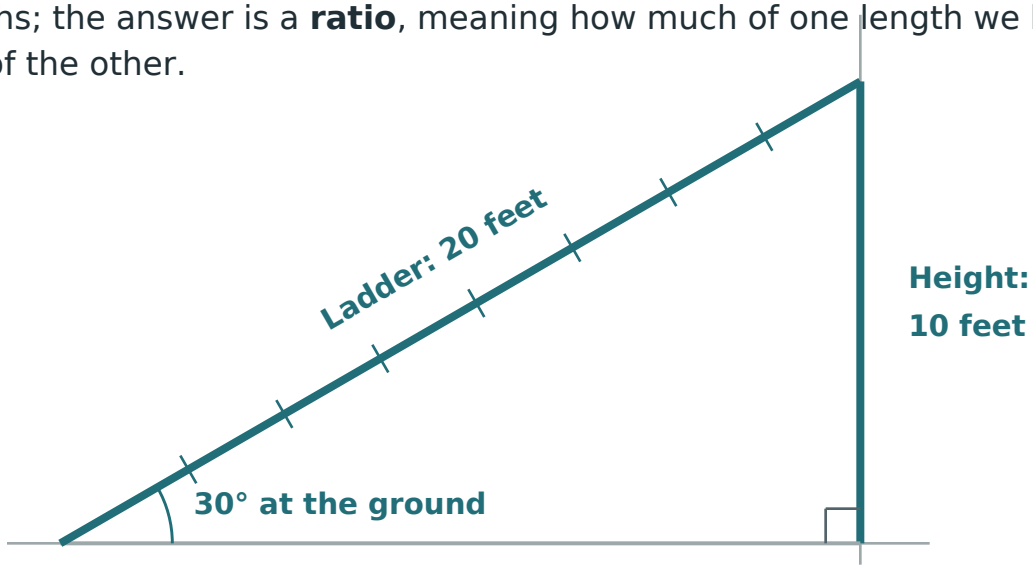
This is a paper example, not advice for safely positioning a real ladder.

QUESTION 1 / SINE — PRONOUNCED “SIGN”

How high does the ladder reach?

We know the ladder is 20 feet long and makes a 30-degree angle with the ground. We want the **height** where it touches the wall.

Sine is the height divided by the ladder length. “Divided by” compares the two lengths; the answer is a **ratio**, meaning how much of one length we have for each unit of the other.



Base: about 17.32 feet

For 30 degrees, the height is always **half the ladder length**. The equal-sided triangle on page 6 explains why; a calculator or table supplies the same ratio.

On a calculator, **sin** means sine, **=** means “equals,” and **0.5** means one half. So **sin(30°) = 0.5**. Below, ÷ means “divided by” and × means “times.”

SINE: HEIGHT DIVIDED BY LADDER LENGTH

Sine = height ÷ ladder length

Our height must be half of 20 feet:

FIND THE HEIGHT

20 feet × 0.5 = 10 feet

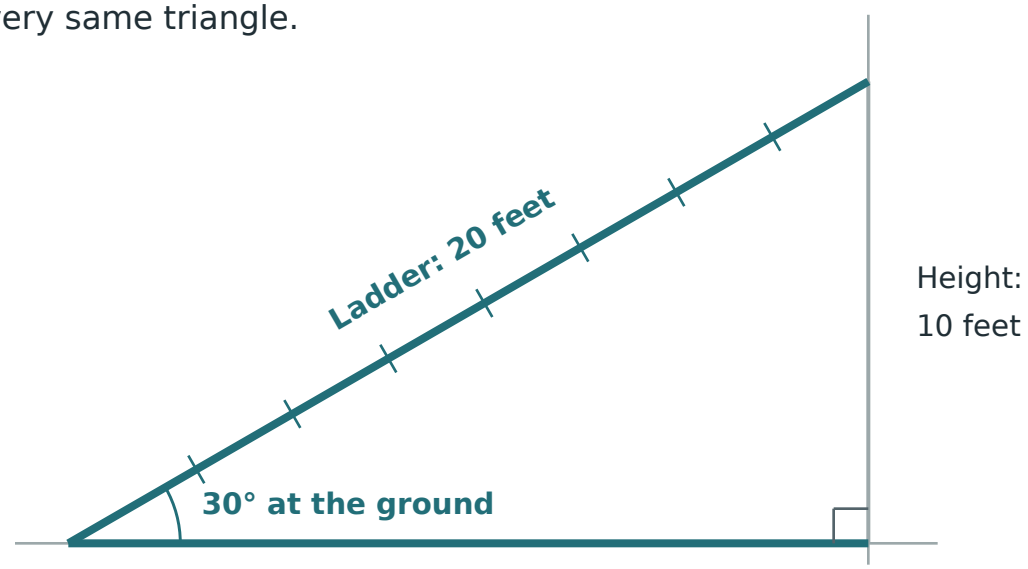
Sine is the **ratio**, not the height itself. Multiplying by the ladder length gives the height.

QUESTION 2 / COSINE — PRONOUNCED “CO-SIGN”

How far is the foot from the wall?

Keep the same ladder and the same ground angle. This time, we want the **base**: the distance along the ground from the foot of the ladder to the wall.

Cosine compares that base with the ladder length. It is a different question about the very same triangle.



For 30 degrees, a calculator gives about **0.866**: the base is about 0.866 times the ladder length. The short name is **cos**.

COSINE: BASE DIVIDED BY LADDER LENGTH

Cosine = base ÷ ladder length

To find the base, multiply the ladder length by that ratio. The symbol ≈ means “approximately equals,” or “about.”

FIND THE BASE

20 feet × 0.866 ≈ 17.32 feet

We use “about” because 0.866 is a rounded calculator answer.

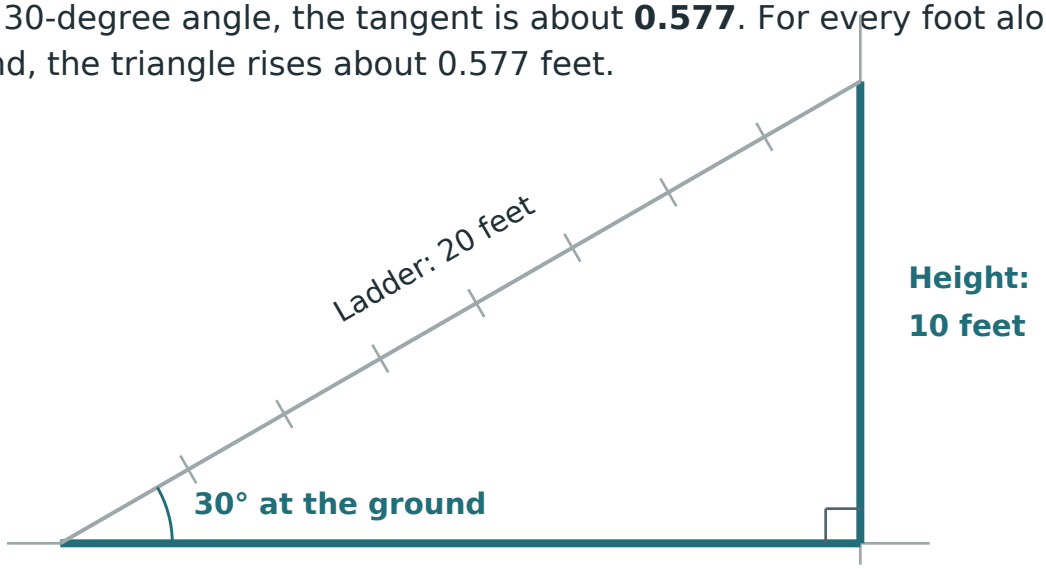
Relative to our ground angle, the height is the **opposite** side: it is across from the angle. The base is the **adjacent** side: it touches the angle and is not the hypotenuse.

QUESTION 3 / TANGENT — SHORTENED TO “TAN”

What if we know the base instead?

Imagine we know the ground angle and the distance from the wall, but not the ladder length. Can we still find the height? Yes. **Tangent** compares the height directly with the base.

For a 30-degree angle, the tangent is about **0.577**. For every foot along the ground, the triangle rises about 0.577 feet.



TANGENT: HEIGHT DIVIDED BY BASE

Tangent = height ÷ base

Our calculator writes this as **tan(30°) ≈ 0.577**. Here “tan” is just the short name of the ratio. It is not another side or another measurement.

FIND THE HEIGHT FROM THE BASE

17.32 feet × 0.577 ≈ 10 feet

Why does this work? A 40-foot ladder at 30 degrees reaches 20 feet high. Double the size; keep the shape. The height-to-ladder ratio stays one half: **10 ÷ 20 = 20 ÷ 40 = 0.5**.

All three ratios work this way: keep the angle and they stay the same, whatever the size. Change the angle and the ratios change.

Calculator tip: select **DEG** (degrees), not RAD (another angle unit, radians). Use sin, cos or tan with 30.

FROM A LADDER TO A LUNAR LANDSCAPE

A shadow makes the same triangle

Now replace the wall height with a mountain’s height. Replace the base with its shadow on level ground. A ray of sunlight takes the place of the sloping ladder.

The **Sun’s elevation angle** means how high the Sun is above the local horizon—the level direction at that place. Here we will use 30 degrees again.

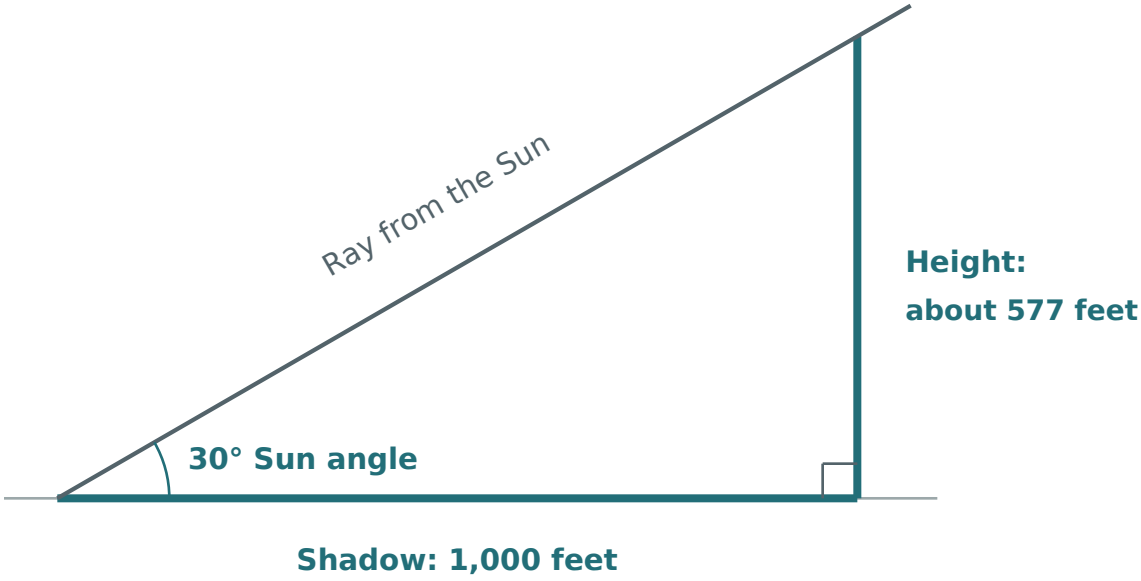


Illustration only: the upright line represents the mountain’s height.

TANGENT STILL MEANS HEIGHT DIVIDED BY BASE

Height = shadow length × tan(Sun angle)

Suppose a shadow is 1,000 feet long and the Sun is 30 degrees up. Tangent tells us the height is about 0.577 times the shadow length.

AN INVENTED EXAMPLE — NOT GALILEO’S DATA

1,000 feet × 0.577 ≈ 577 feet

This simple method also works on the Moon, approximately. We need the **true ground length** of the shadow, not an uncorrected length in a photograph, and the Sun’s angle at that lunar location.

We assume level ground and sunlight traveling in the same direction. Real measurements may need corrections for slopes, lunar curvature and the viewing angle. [2]

GALILEO’S MOON / WHAT HE ACTUALLY DID

The light reached the peaks first

In 1610, Galileo described bright mountaintops shining inside the Moon’s dark region. Picture sunrise on Earth: a high peak can catch the Sun while the valley below is still dark.

He measured how far those bright peaks lay beyond the **terminator**—the boundary between lunar day and night. He compared that distance with the Moon’s diameter, meaning its full width through the center.

His published calculation used the Moon’s curved shape and a right triangle. He used **Pythagoras’s rule** (Section 2, page 5), the relationship between a right triangle’s sides, to work out how far a peak had to rise to reach the sunlight.

That is not the shadow-times-tangent calculation on the previous page. Both turn light and triangles into a height, but they are different methods.

Galileo reported a height **“more than 4 Italian miles.”** Those are historical Italian miles, not automatically four of today’s English miles. This was his estimate—not a modern survey. [1]

TAKE THIS ONE IDEA WITH YOU

An angle fixes the triangle’s proportions. One known length then tells us the scale. Each function answers a different question about the **same ladder**:

Sine: height compared with ladder length.
Cosine: base compared with ladder length.
Tangent: height compared with base.

The names are new. The idea is simply comparing lengths.

Where the story goes next

Geometry gives us a shape; trigonometry compares its sides. What if the shape or position changes from moment to moment? That is our next subject: Section 4, calculus (page 13).

Galileo’s original text and the related shadow method are linked in the sources on page 22. The shadow example on page 11 is invented, not Galileo’s data.

4.1 / NEWTON, LEIBNIZ AND THE MOVING WORLD

Now let something change

The shelter total, triangle and ladder all held still while we examined them. But a truck speeds up, a planet changes direction, and a tide rises. **Calculus** helps us describe continuous change and what builds up from it.

In the late 1600s, Isaac Newton and Gottfried Wilhelm Leibniz developed calculus independently, building on earlier work. Newton connected this new mathematics closely with motion. It was not one person’s invention out of nothing.

How fast, right now?

A rate of change

How much altogether?

Accumulation

A **rate of change** compares a change in one quantity with a change in another. Speed is a familiar example: how much the distance changes as time passes.

Accumulation means adding up what happens over an interval. If we know the truck’s speed throughout a journey, we can add up the small distances it covers to find the whole trip.

These two questions are partners: from the journey’s distance record, find the changing speed; from the changing speed, recover the distance traveled.

The earlier sections are foundations, not a claim that Pythagoras invented calculus. Geometry describes positions and lengths; algebra expresses relationships. Calculus asks how those relationships behave as things change.

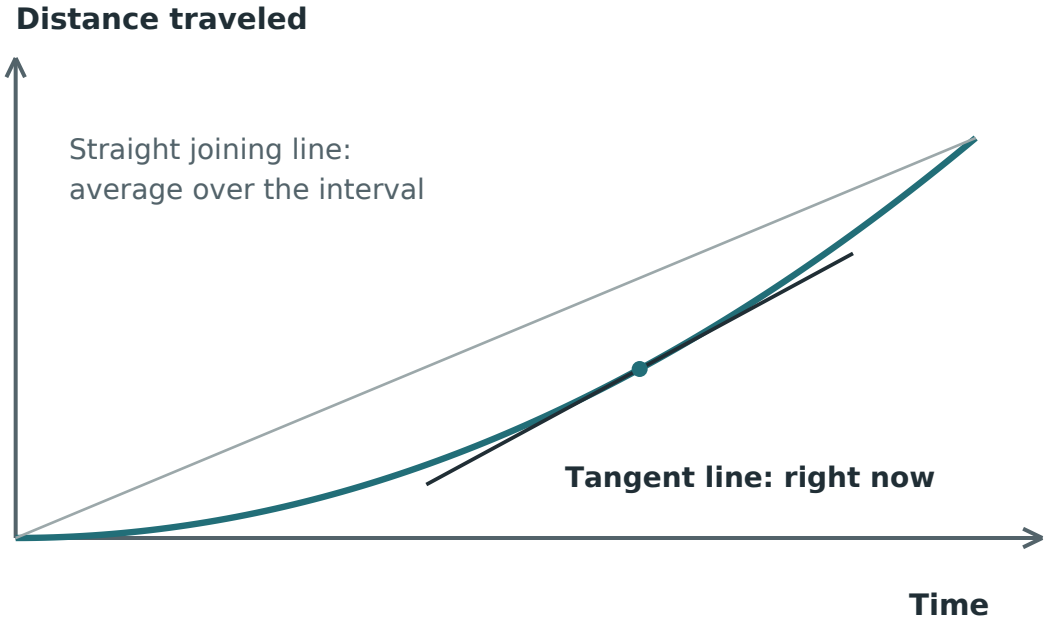
You can understand the central ideas without learning the formal symbols first. We will stay with the truck before returning to the sky.

4.2 / AVERAGE IS NOT THE SAME AS RIGHT NOW

A speedometer and a curve

Suppose the truck travels 60 miles in two hours. Its **average speed** is the total distance divided by the total time: **60 ÷ 2 = 30 miles per hour**. That does not mean it moved at 30 the whole way.

Instantaneous speed means speed at a particular moment—the kind of reading a speedometer gives. To estimate it, measure distance over a very short time. Calculus asks what happens as that time interval gets smaller and smaller.



This picture is a **graph**: time runs across; distance runs upward. The curve records a journey. Its **slope** means rise compared with horizontal run—here, distance change compared with time change.

A **tangent line** follows the curve’s direction at one point. Its slope gives the rate there. Finding that local rate is called **differentiation**; the rate we find is called a **derivative**.

This is not quite the same use of “tangent” as on page 10. There, **tan** is a triangle’s height-to-base ratio. Here, a **tangent line** is a line beside a curve. Their connection is slope: rise divided by run.

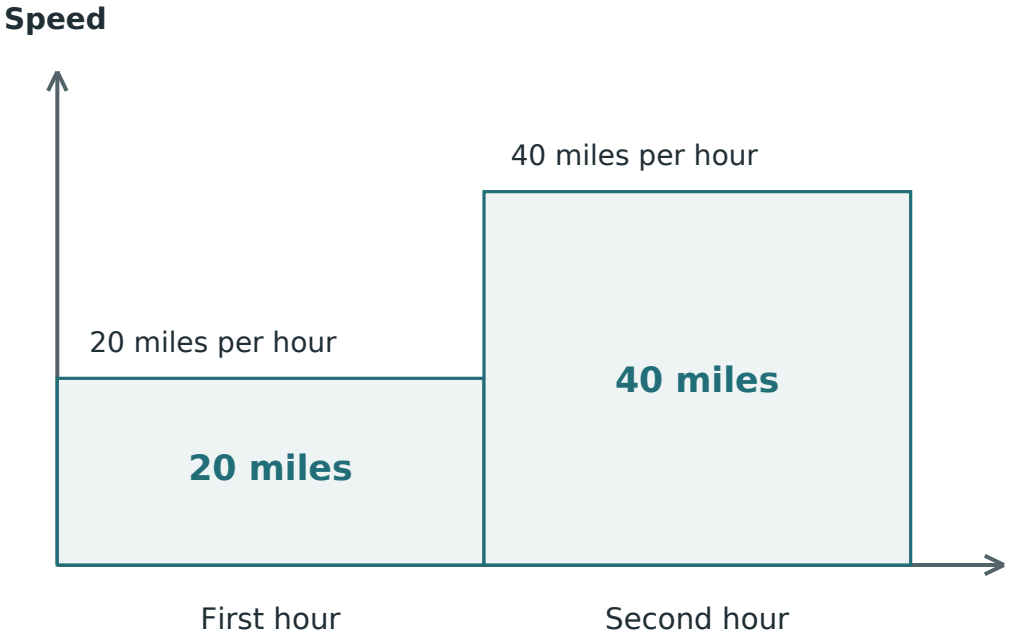
The graph is conceptual, not the numerical record of the 60-mile trip above. No formula for the curve is needed to read the idea.

4.3 / ADD THE LITTLE PIECES

From speed back to distance

Now turn the question around. Suppose we know how fast the truck moved. If it held 20 miles per hour for one hour, it covered 20 miles. If it then held 40 for another hour, it covered another 40.

So the whole distance is **20 + 40 = 60 miles**. We found a total by adding the contributions of different parts of the journey.



In a speed-versus-time graph, a rectangle’s height is the speed and its width is the time. Multiplying them gives the distance during that time. The **area under the speed graph** therefore represents distance.

When speed changes smoothly, divide the journey into short time pieces. Use each piece’s speed to estimate its distance, then add the distances.

Calculus makes this idea precise by taking smaller and smaller pieces. Adding a continuously changing quantity this way is called **integration**; the accumulated result is an **integral**.

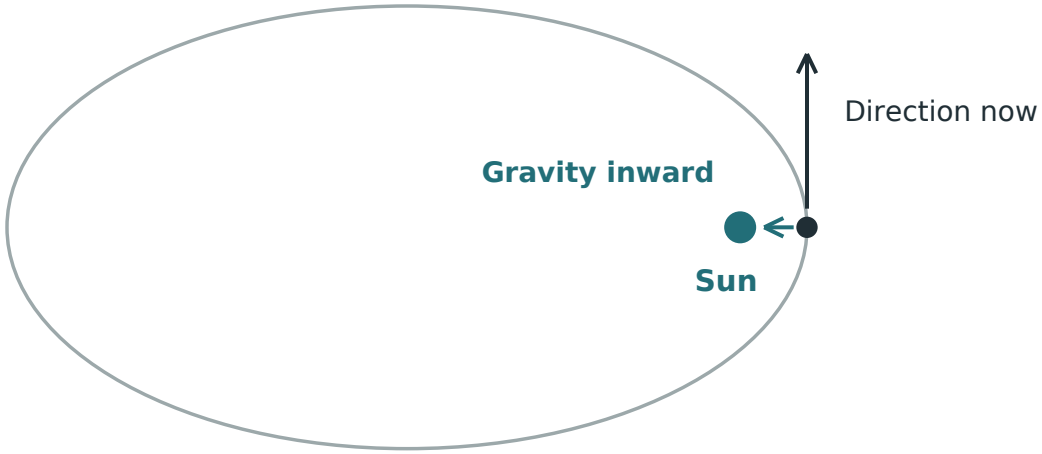
Differentiation asks for the local rate; integration adds the pieces. Under suitable conditions, these are reverse operations. That partnership is the central link in calculus.

4.4 / FROM MOVING TRUCKS TO MOVING PLANETS

Orbits, tides and changing direction

For Newton, motion raised a deeper question: how does a pull change an object’s journey? **Gravity** is the attraction between masses—between the Sun and a planet, or between Earth and the Moon.

Velocity means speed with a direction. **Acceleration** means a change in velocity, so turning counts even when speed stays the same. Calculus relates position, velocity and acceleration from moment to moment.



Schematic ellipse; sizes and distances are not to scale.

Forward motion and the Sun’s inward pull make the planet’s path curve. Without gravity it would continue straight in its current direction. Orbits need not be circles; many are **ellipses**—oval-shaped paths.

The Moon pulls harder on Earth’s near side than on its center, and less on the far side. This **difference across Earth** helps stretch the oceans and creates tides. The Sun contributes too.

Newton’s gravity explains the basic tide-raising forces. Local water-level predictions also need ocean depth, coastline shape and other effects. Calculus describes change; it does not make geography disappear.

Geometry describes shapes; calculus describes how they change.

4.5 / DIFFERENTIAL EQUATIONS, IN EVERYDAY LIFE

Why hot coffee cools more slowly

A fresh cup of coffee cools quickly at first, then more slowly as it approaches room temperature. A **differential equation** connects a changing quantity with its **rate of change**—how quickly it changes. Here is a simple model, called Newton's law of cooling.

First, name the pieces. **T** is the coffee's temperature, in degrees Fahrenheit. **t** is elapsed time, in minutes. **T_room** is the steady room temperature; "room" is a label, not multiplication. The **gap** is coffee temperature minus room temperature: **T − T_room**. The sign **−** means subtract.

dT/dt reads "rate of temperature change": degrees Fahrenheit per minute. Think of **dT** as a tiny temperature change and **dt** as a tiny time change; the derivative describes the limiting rate. **k** is a positive **cooling coefficient**—a multiplier telling how quickly the gap shrinks. Its units are **per minute**, or inverse time.

The equals sign means both sides have the same value. Parentheses group the gap. Writing k beside parentheses means multiply. Now we can read the equation:

dT/dt = −k(T − T_room)

In words: **the size of the cooling rate is k times the gap**. The minus sign before k makes the temperature-change rate negative: hot coffee's temperature is falling. A smaller gap gives slower cooling.

The **time constant** is one characteristic cooling time. Its name is **tau**, written **τ**. The rule is **τ = 1/k**: one divided by k. Because k is per minute, tau is in minutes. The actual time depends on the cup and conditions; we have not chosen a number of minutes.

The sign **%** means "out of a hundred." After **one** time constant, about **37% of the original gap remains**: 63% has closed. After **three**, about **5% remains**: 95% has closed. These percentages describe the **gap**, not the room temperature.

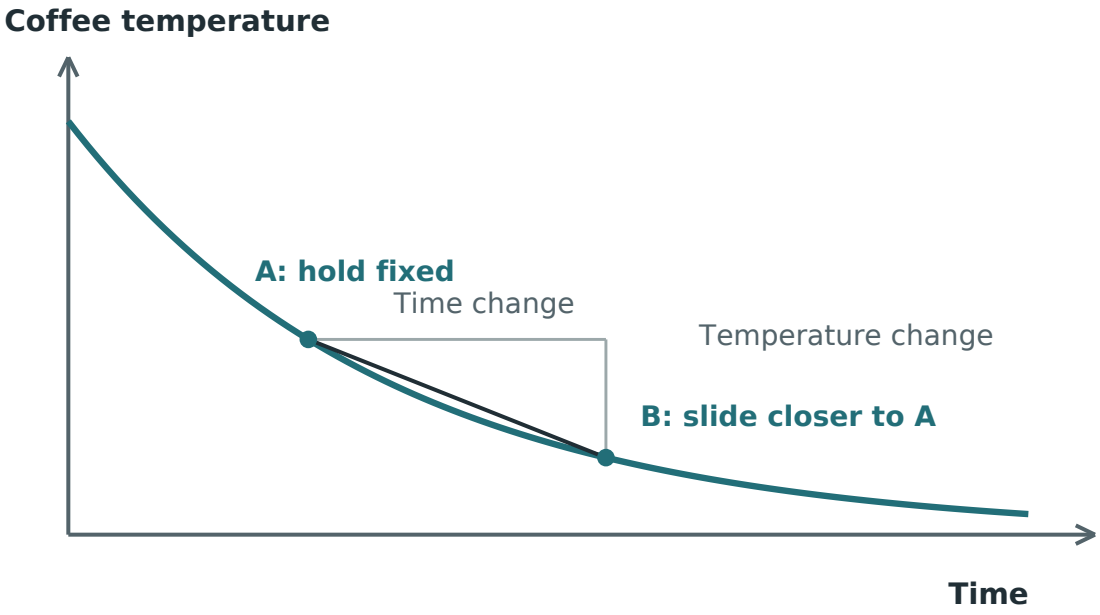
Example: coffee starts at 170°F in a 70°F room: a 100-degree gap. "°F" means degrees Fahrenheit. After one time constant it is about **107°F**; after three, about **75°F**.

The model assumes steady room temperature and approximately constant k. It only approximates real coffee. It approaches room temperature without reaching it exactly in finite time: page 14's local-rate idea at work.

4.6 / THE DERIVATIVE, ONE INSTANT AT A TIME

Two points, then one instant

On a **coffee-temperature graph**, time runs horizontally and temperature runs vertically. Choose an instant and mark its temperature as **point A**. Choose a nearby later instant and mark its temperature as **point B**. The letters are only labels.



Schematic curve, not observed measurements. The dark line joins A and B.

Join the points. The line’s slope is the **vertical temperature change divided by the horizontal time change**—rise divided by run. The run is positive; the rise is negative because coffee cools. The ratio is degrees Fahrenheit per minute.

Hold A fixed and slide B closer along the curve. Each two-point ratio estimates the slope at A. The value these ratios approach as the time gap shrinks is the **limit**: the derivative at A. We never divide by zero.

Every instant has its own slope. In this cooling model the slopes become less negative as the curve flattens. The limit is exact for the ideal smooth model; real measurements only estimate it.

5 / SIX PATTERNS, WITH EVERYDAY EXAMPLES

Growth rates: steady and square

5.1 Linear — add the same amount

Linear growth adds the same amount for every equal step in the input.

Example: An odometer records 10 more miles each hour while a vehicle holds 10 miles per hour.

Driving time (hours)	Trip distance (miles)
1	10
2	20
3	30
4	40
5	50
6	60
7	70
8	80

5.2 Quadratic — the square pattern

Quadratic growth follows the area of a square as its side gets longer.

Example: A square garden bed 2 feet on each side covers 4 square feet; a square foot is a 1-foot square.

Garden bed’s side (feet)	Garden area (square feet)
1	1
2	4
3	9
4	16
5	25
6	36
7	49
8	64

“Input” is the number we start with; “output” is what the rule gives back. These tables use different units, so compare their patterns—not miles directly with square feet.

5 / SIX PATTERNS, WITH EVERYDAY EXAMPLES

Growth rates: cubes and doubling

5.3 Cubic — the cube pattern

Cubic growth follows the volume of a cube as its side gets longer.

Example: A concrete cube 2 feet on each side has 8 cubic feet of volume; a cubic foot is a 1-foot cube.

Cube’s side (feet)	Cube’s volume (cubic feet)
1	1
2	8
3	27
4	64
5	125
6	216
7	343
8	512

5.4 Exponential — multiply each time

Exponential growth multiplies by the same factor at each equal step.

Example: Imagine bacteria doubling each hour, starting with one at hour zero.

Time elapsed (hours)	Bacteria in the ideal model
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256

Bacteria cannot double forever: this model assumes unlimited resources. Doubling may start below a square pattern before eventually passing it.

5 / SIX PATTERNS, WITH EVERYDAY EXAMPLES

Growth rates: orders and digits

5.5 Factorial — count possible orders

Factorial growth counts the different ways to place distinct people in a row of seats.

Example: Three different people can occupy a row of three seats in six different ways.

n! (“n factorial”) multiplies whole numbers 1 through group size n; ! is not excitement.

People to seat (n)	Seating arrangements (factorial)
1	1
2	2
3	6
4	24
5	120
6	720
7	5,040
8	40,320

5.6 Logarithmic — count multiplication steps

Logarithmic growth adds a fixed amount whenever the input grows by a fixed factor.

Example: log10 counts tenfold steps from 1: each step multiplies by ten.

Inputs are powers of ten, not 1–8; “digits” counts written figures (100 has three).

Input number	Tenfold steps (log10)	Written digits
1	0	1
10	1	2
100	2	3
1,000	3	4
10,000	4	5
100,000	5	6
1,000,000	6	7
10,000,000	7	8

Digit counts change in whole-number steps, not along a smooth log curve. “Log10 plus one” gives the digit counts shown **for these powers of ten**.

For these fixed rules, factorial eventually grows fastest; small-input rankings differ. Compare patterns, not quantities with different units.

READ ONLY IF YOU WANT TO FOLLOW THE THREADS

Sources, and what to remember

The historical stories are real; the shelter, truck and shadow numbers are teaching examples. The master paper reworks the standalone ladder lesson into Section 3. Original PDFs are left unchanged.

Geometry and its history
University of St Andrews, MacTutor: Euclid and Pythagoras. Euclid organized earlier knowledge; the right-triangle relationship predates Pythagoras.

[1] Galileo's actual mountain calculation
Galileo, Sidereus Nuncius (1610), E. S. Carlos translation (1880), pp. 28–30: Project Gutenberg ebook 46036. Sunlit peaks beyond the terminator; right-triangle calculation; "more than 4 Italian miles."

[2] The related shadow method
Mary McIntyre, BBC Sky at Night Magazine: "How to measure craters on the Moon". Height equals shadow length times the tangent of the local Sun angle, with simplifying assumptions.

Calculus and its development
University of St Andrews, MacTutor: "A history of the calculus". Earlier work, Newton, Leibniz, rates and accumulation. Orbits: NASA, "Gravity & Mechanics".

Why tides happen
NOAA: "Tides" and "What Causes Tides?". Uneven gravitational pull, the Moon and Sun, and the influence of local geography.

Four questions, one growing toolkit
Algebra: What number is missing?
Geometry: What shape and lengths do we have?
Trigonometry: What do the angles tell us about the sides?
Calculus: How is it changing, and what does it add up to?

You do not have to memorize every word. Recognizing the question is already a beginning.